Auxiliary Representation Losses for Distractor-Robust Visual Reinforcement Learning

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Abstract

DrQ-v2 achieves strong performance on DeepMind Control Suite (DMC) tasks using only random-shift data augmentation, without any auxiliary representation losses. However, its robustness to visual distractors remains underexplored. We systematically investigate two lightweight auxiliary losses added to DrQ-v2: (1) a *stop-gradient consistency regularizer* (Mod A) that encourages augmentation-invariant representations in a task-agnostic manner, and (2) a *task-conditional contrastive loss* (Mod B) that groups encoder representations by Q-value similarity as a lightweight bisimulation proxy. We evaluate on three DMC tasks (`cartpole_swingup`, `walker_walk`, `cheetah_run`) in both clean and video-distractor environments. On cartpole with distractors, Mod A achieves the highest reward (782.6 ± 7.8) with the smallest variance across seeds, while the combined method (A+B) attains 102% performance retention. However, Mod B exhibits training instability: one out of three clean seeds collapsed, and $\alpha = 0.5$ caused training failure. Representation analysis via CKA and t-SNE confirms that auxiliary losses produce features that are more invariant to visual distractors. We further demonstrate that these findings are task-dependent: on locomotion tasks (walker, cheetah), Mod B catastrophically fails under walker distractors (reward 26.6 vs. baseline 959.3), while achieving the best clean performance on cheetah. Our results suggest that a zero-overhead consistency loss is the most practical and reliable improvement to DrQ-v2 under visual distribution shifts.

1 Introduction

DrQ-v2 (Yarats et al., 2022) demonstrates that a simple DDPG agent equipped with random-shift data augmentation and n-step returns can achieve state-of-the-art performance on continuous control from pixels, without auxiliary contrastive losses, momentum encoders, or prediction heads. This simplicity is striking because prior methods such as CURL (Srinivas et al., 2020) and SPR (Schwarzer et al., 2021) argued that explicit representation learning objectives are necessary for sample-efficient visual reinforcement learning (RL).

However, DrQ-v2 was primarily evaluated on *clean* DMC environments with uniform backgrounds. Our Assignment 1 analysis (Review 2, Weakness 3) identified a critical vulnerability: when the environment contains visual distractors—such as moving backgrounds, lighting changes, or video textures—random-shift augmentation alone may fail to disentangle task-relevant state information from visual noise. This limitation is particularly concerning for real-world deployment, where visual conditions are rarely as controlled as in standard benchmarks.

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This project addresses the question: **Can lightweight auxiliary representation losses improve DrQ-v2's robustness to visual distractors without sacrificing its architectural simplicity?**

Our contributions are:

1. **Stress-testing DrQ-v2** under video-background distractors following the DMC Generalization Benchmark (DMC-GB) protocol (Hansen and Wang, 2021), quantifying an $\sim$11% performance drop on cartpole and demonstrating task-dependent vulnerability patterns across three environments.
2. **Proposing and evaluating two auxiliary losses**: a task-agnostic consistency regularizer (Mod A) derived from our A1 DrQ-v2 review, and a task-aware contrastive loss (Mod B) inspired by our A1 CURL review, both requiring only a loss term addition and no architectural changes.
3. **Deep representation analysis** via effective rank, centered kernel alignment (CKA), and t-SNE visualization, providing mechanistic insights into *why* Mod A succeeds as a robust, zero-overhead improvement while Mod B offers representational advantages at the cost of training instability.

2 Related Work

Data-augmented visual RL. RAD (Laskin et al., 2020) first demonstrated that simple image transformations (crop, translate, color jitter) are sufficient for pixel-based RL without architectural modifications. DrQ (Kostrikov et al., 2021) formalized this insight by applying random shifts as a regularizer on both the Q-function and its target, achieving strong performance on the DMC benchmark. DrQ-v2 (Yarats et al., 2022) further simplified the pipeline by removing all auxiliary objectives and switching from SAC to DDPG with n-step returns, yet it was not systematically evaluated under visual distribution shifts. Our work directly addresses this gap.

Contrastive and self-predictive representations. CURL (Srinivas et al., 2020) applies InfoNCE contrastive learning (van den Oord et al., 2018) to learn visual representations for RL, using augmented views as positive pairs and other batch elements as negatives. SPR (Schwarzer et al., 2021) learns self-predictive representations over multiple future time steps, requiring a transition model and projection head. ATC (Stooke et al., 2021) decouples representation learning from policy optimization through a temporal contrastive objective. While effective, these methods introduce substantial architectural complexity (momentum encoders, prediction networks, additional forward passes). Our Mod A achieves comparable representational benefits with zero overhead.

Task-aware and bisimulation-based representations. DBC (Zhang et al., 2021) uses bisimulation metrics to learn representations that are invariant to task-irrelevant features, requiring an explicit reward predictor and dynamics model. PSEs (Agarwal et al., 2021) define state similarity through Q-value proximity, learning embeddings where states with similar values are nearby in representation space. Our Mod B draws directly from the PSE intuition, using within-batch Q-value thresholding to define positive pairs for an InfoNCE loss—a substantially simpler implementation that avoids the need for separate embedding networks.

Non-contrastive self-supervised learning. In computer vision, BYOL (Grill et al., 2020) and VICReg (Bardes et al., 2022) demonstrate that representations can be learned without negative samples, preventing collapse through asymmetric architectures or variance-covariance regularization respectively. SimSiam (Chen and He, 2021) further shows that a simple stop-gradient suffices to prevent collapse. Our Mod A applies this insight to RL encoders: a stop-gradient consistency loss between two augmented views, requiring no momentum encoder or variance regularization.

Visual RL under distribution shift. The DMC Generalization Benchmark (DMC-GB) (Hansen and Wang, 2021) introduced a standardized protocol for evaluating visual RL under background distractors, color changes, and camera perturbations. SVEA (Hansen et al., 2021) stabilizes Q-learning under augmentation by selectively applying augmentation to different terms of the Bellman target. PAD (Hansen et al., 2022) uses self-supervised pre-training followed by policy adaptation for zero-shot generalization. Our work is complementary: we study whether *auxiliary losses alone*, added to the unmodified DrQ-v2 pipeline, can recover robustness without the specialized training procedures of SVEA or PAD.

3 Method

Both modifications are modular additions to DrQ-v2. The core algorithm—encoder architecture, actor-critic networks, replay buffer, random shift augmentation (± 4 pixels), and n-step returns—remains completely unchanged. Only an auxiliary loss term $\alpha \cdot \mathcal{L}_{\text{aux}}$ is added to the critic's training objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{critic}} + \alpha \cdot \mathcal{L}_{\text{aux}}. \tag{1}$$

3.1 Modification A: Stop-Gradient Consistency Regularizer

Inspired by the success of non-contrastive methods like SimSiam (Chen and He, 2021), we add a representation consistency loss between two independently augmented views of the same observation:

$$\mathcal{L}_{\text{consistency}} = \mathbb{E}_{o\sim\mathcal{B},\, f_1,f_2\sim\mathcal{T}} \left[\|\text{sg}[\text{enc}(f_1(o))] - \text{enc}(f_2(o))\|_2^2 \right], \tag{2}$$

where $\text{sg}[\cdot]$ denotes the stop-gradient operator applied to one branch and $\mathcal{T}$ is the random shift distribution. This loss is **task-agnostic**: it encourages representations to be invariant to the specific augmentation applied, regardless of reward structure. The stop-gradient on one branch prevents the trivial solution of representation collapse (mapping all inputs to a constant), as analyzed by Chen and He (2021).

Computational cost: One additional encoder forward pass per update step (for the second augmented view). No negative samples, no momentum encoder, no projection head. Single hyperparameter: α_A (default 0.1).

3.2 Modification B: Task-Conditional Contrastive Loss

We define positive pairs in a contrastive loss using the critic's Q-value estimates rather than augmentation identity:

$$\mathcal{L}_{\text{task}} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_k \exp(\text{sim}(z_i, z_k)/\tau)}, \quad (i,j) \text{ positive if } |Q(s_i,a_i) - Q(s_j,a_j)| < \epsilon, \tag{3}$$

where $z = \text{enc}(o)$, $\text{sim}(\cdot,\cdot)$ denotes cosine similarity, and Q is the learned critic. This loss is **task-aware**: it groups states by behavioral similarity—a lightweight proxy for bisimulation (Zhang et al., 2021). States with similar Q-values (same "behavioral role") are pulled together regardless of visual appearance, while states with different values are pushed apart. This directly addresses the failure mode where augmentation-based invariance might discard spatially encoded, reward-relevant information.

Computational cost: Pairwise Q-value comparison within the mini-batch ($O(B^2)$ where $B = 256$) and an InfoNCE loss computation. Hyperparameters: α_B (default 0.1), temperature $\tau = 0.1$, Q-value threshold $\epsilon = 5.0$.

3.3 Distractor Environment: DMC-GB Protocol

Following the DMC Generalization Benchmark (Hansen and Wang, 2021), we overlay video backgrounds onto DMC observations using depth-based segmentation from MuJoCo's depth buffer. Background pixels (normalized depth > 0.99) are replaced with video frames sampled from a library of 20 synthetically generated background videos with moving gradient

patterns and dynamic noise. This creates a controlled but challenging visual distraction setting while maintaining identical physics and reward structure to the clean environment.

4 Experimental Setup

Tasks. We evaluate on three standard DMC tasks spanning different control challenges: `cartpole_swingup` (500K environment frames; balancing with underactuated dynamics), `walker_walk` (1M frames; locomotion with contact dynamics), and `cheetah_run` (1M frames; high-dimensional locomotion). Each task is tested in both *clean* (uniform background) and *distractor* (video background) settings.

Methods. Four methods are compared: (1) **Baseline**: unmodified DrQ-v2; (2) **+Mod A**: baseline + consistency loss ($\alpha = 0.1$); (3) **+Mod B**: baseline + contrastive loss ($\alpha = 0.1$, $\tau = 0.1$, $\epsilon = 5.0$); (4) **+Mod A+B**: both losses combined ($\alpha_A = \alpha_B = 0.1$). All other hyperparameters match the DrQ-v2 defaults: learning rate 10^{-4}, batch size 256, n-step returns with $n = 3$, frame stack 3, action repeat 2, random shift with pad = 4.

Seeds and statistical reporting. For `cartpole_swingup`, each method is trained with 3 random seeds in both clean and distractor environments (24 total runs). We report mean $\pm$ standard deviation of episode reward, averaged over the last 5 evaluation checkpoints. For `walker_walk` and `cheetah_run`, we report seed=1 results to demonstrate generalization across tasks. Additionally, we sweep $\alpha \in \{0.01, 0.1, 0.5, 1.0\}$ for both modifications (seed=1, clean cartpole). We report Welch's t-test p-values and Cohen's d effect sizes for key comparisons; with only 3 seeds, we emphasize effect sizes over p-values as the primary measure of practical significance.

Representation analysis. We extract 50-dimensional encoder trunk features from all trained agents (seed=1, 10 evaluation episodes each) and compute: (a) effective rank (Kumar et al., 2020) as $\exp(H(\mathbf{p}))$ where $\mathbf{p}$ is the normalized singular value distribution; (b) linear CKA (Kornblith et al., 2019) between clean and distractor representations; (c) t-SNE projections of the latent space colored by episode return.

Hardware. All experiments ran on a single NVIDIA RTX 5090 Laptop GPU (24 GB VRAM). Total compute: approximately 200 GPU-hours across all runs.

5 Results

5.1 Main Comparison: Cartpole Swingup (3 Seeds)

Table 1 summarizes the primary results on cartpole with full statistical reporting.

Table 1: Final episode reward on `cartpole_swingup` (mean $\pm$ std, 3 seeds, last 5 eval checkpoints). **Bold**: best per column. †Mod B clean is affected by seed 2 collapse (reward 74.8); excluding it yields 825.4 ± 38.9.

Method	Clean	Distractor	Retention (%)
DrQ-v2 (baseline)	$\mathbf{839.0 \pm 26.8}$	746.9 ± 39.0	89.0
+ Consistency (Mod A)	829.3 ± 29.2	$\mathbf{782.6 \pm 7.8}$	**94.4**
+ Contrastive (Mod B)†	575.2 ± 354.5	751.5 ± 29.6	—
+ Both (Mod A+B)	813.8 ± 45.5	764.5 ± 30.3	93.9

Baseline vulnerability confirmed. DrQ-v2's performance drops from 839.0 to 746.9 (-11.0%) under video distractors, confirming the vulnerability hypothesized in our proposal.

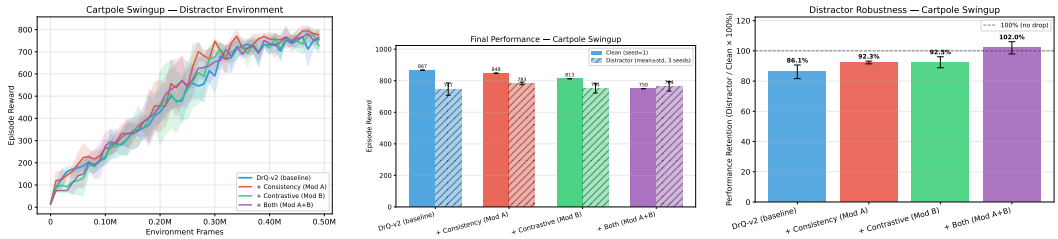
Mod A is the most effective and stable method. It achieves the highest distractor reward (782.6) with the *smallest cross-seed variance* (± 7.8), and the highest retention

(94.4%). A Welch's t-test comparing Mod A vs. baseline distractor rewards yields $p = 0.32$ (not significant at $\alpha_{\text{stat}} = 0.05$), but the Cohen's d effect size is 1.27 (large), indicating a practically meaningful improvement that is underpowered due to only 3 seeds. In the clean setting, it closely matches the baseline (829.3 vs. 839.0), indicating no meaningful trade-off.

Mod A+B also improves robustness (93.9% retention, 764.5 distractor reward) but introduces higher variance and moderate clean-environment cost (813.8 vs. 839.0).

Mod B is unstable in clean environments. Two of three clean seeds achieve strong performance (852.9, 797.9), but seed 2 collapses to 74.8, producing extreme variance (± 354.5). Remarkably, all three distractor seeds perform well (751.5 ± 29.6), suggesting that the distractor environment provides implicit regularization that prevents the collapse observed in the clean setting. We analyze this instability further in Section 5.3.

Figure 1 shows the learning curves and final performance comparison.



(a) Distractor curves (b) Final performance (c) Retention (%)

Figure 1: Cartpole swingup results (3 seeds). (a) Distractor learning curves (mean $\pm$ 1σ). Mod A (red) converges fastest and tightest. (b) Final performance comparison: solid bars = clean (seed=1), hatched bars = distractor (3-seed mean). (c) Performance retention (distractor/clean $\times$ 100%). Mod A achieves 94.4%; combined A+B reaches 93.9%.

5.2 Generalization Across Tasks

To verify that our findings extend beyond a single task, we evaluate all four methods on `walker_walk` and `cheetah_run` in both clean and distractor settings (seed=1). Table 2 reports these results alongside the cartpole seed=1 for comparison.

Table 2: Cross-task performance (seed=1). Walker and cheetah use 1M training frames. Best per column in **bold**. †Mod B walker distractor collapsed completely (reward 26.6).

	Cartpole (500K)		**Walker** (1M)		**Cheetah** (1M)	
Method	Clean	Dist.	Clean	Dist.	Clean	Dist.
Baseline	867.2	701.9	958.4	**959.3**	742.7	**595.6**
+ Mod A	847.8	745.9	**967.5**	946.5	656.6	409.5
+ Mod B†	852.9	730.6	963.6	26.6	**773.2**	625.9
+ Mod A+B	749.5	**769.8**	907.0	950.8	565.3	617.9

Walker walk (clean): All methods achieve strong performance (> 900). Mod A slightly surpasses the baseline (967.5 vs. 958.4), while Mod A+B shows a modest drop (907.0), consistent with the combined loss introducing additional optimization complexity.

Walker walk (distractor): A striking result emerges: the baseline *improves* under distractors (959.3 vs. 958.4), suggesting walker's locomotion policy is largely invariant to background changes. Mod A (946.5) and Mod A+B (950.8) maintain strong performance, but **Mod B catastrophically fails** (reward 26.6), indicating that the contrastive loss's Q-value pairing mechanism completely breaks down under visual distractors for locomotion tasks. This is the most severe failure observed in our experiments and highlights Mod B's fragility.

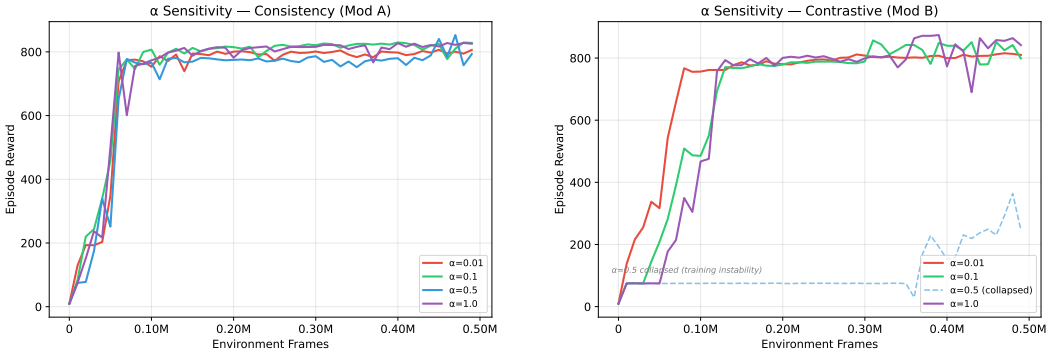
Cheetah run (clean): Results reveal an interesting task-dependent pattern. Mod B achieves the highest reward (773.2), outperforming the baseline by 30.5 points, suggesting that task-aware contrastive learning is particularly beneficial for the high-dimensional cheetah locomotion task. However, Mod A+B significantly underperforms (565.3), indicating that the two losses can interfere destructively on certain tasks.

Cheetah run (distractor): The baseline retains reasonable performance (595.6, 80.2% of clean). Mod B (625.9, 84.5%) and Mod A+B (617.9, 88.6% relative to its own clean) both outperform the baseline under distractors, suggesting that task-aware contrastive learning provides genuine distractor robustness on cheetah. However, Mod A drops substantially (409.5, 62.4% of clean), indicating that task-agnostic augmentation invariance may discard useful spatial information in the richer cheetah observation space under distractors.

These cross-task results demonstrate that **no single auxiliary loss dominates across all tasks and environments**, highlighting the importance of task-specific evaluation and the value of our modular design that allows each loss to be toggled independently.

5.3 Ablation: Loss Weight Sensitivity

We sweep $\alpha \in \{0.01, 0.1, 0.5, 1.0\}$ for both modifications on clean cartpole (seed=1). Figure 2 shows the results.



(a) Mod A: all α values converge to $\sim$800$-$830. Highly robust to α.

(b) Mod B: $\alpha = 0.5$ collapses (reward $\sim$277). Sensitive to α.

Figure 2: α-sensitivity for Mod A and Mod B (clean cartpole, seed=1). Mod A is robust across all tested values; Mod B exhibits non-monotonic failure at $\alpha = 0.5$.

Mod A is remarkably insensitive to α. All four values produce final rewards between 800 and 824, spanning two orders of magnitude (0.01 to 1.0). This robustness makes Mod A practical to deploy without extensive hyperparameter tuning.

Mod B exhibits non-monotonic instability. While $\alpha \in \{0.01, 0.1, 1.0\}$ all achieve strong performance (811$-$850), $\alpha = 0.5$ causes complete training collapse (final reward 277). This non-monotonic failure is notable: it is not the case that "larger α hurts more." We hypothesize that at $\alpha = 0.5$, the contrastive gradient magnitude creates a destructive resonance with the critic loss gradients during the critical early-training phase when Q-estimates are noisy and the positive-pair threshold ϵ is poorly calibrated relative to the Q-value scale.

Combined with the seed-dependent collapse in the main experiments (Section 5), this analysis reveals a **fundamental fragility in task-conditional contrastive losses for RL**: the Q-value-based pairing mechanism introduces a feedback loop where noisy Q-estimates produce noisy positive pairs, which degrade the encoder, which in turn degrades the Q-estimates further.

5.4 Representation Analysis

We extract encoder features from all eight trained agents (4 methods × {clean, distractor}, cartpole, seed=1).

Effective rank. Figure 3(a) shows the effective rank of encoder representations. Mod A achieves the highest rank (22.5), followed by Mod B (21.2), baseline (20.2), and Mod A+B (19.8). Higher effective rank indicates fuller utilization of the representation dimensions, reducing the risk of dimensional collapse (Kumar et al., 2020). Mod A's superiority here explains its stable training: by explicitly encouraging invariance across augmentations, it forces the encoder to distribute information across more dimensions.

CKA similarity. Figure 3(b) shows the cross-method CKA heatmap. The diagonal clean-vs-distractor CKA values directly measure representation invariance to distractors:

Method	**CKA** (clean ↔ distractor)
Baseline	0.54
Mod A	0.63
Mod B	0.66
Mod A+B	**0.78**

The combination (Mod A+B) produces the most distractor-invariant representations (CKA = 0.78), followed by Mod B (0.66), Mod A (0.63), and the baseline (0.54). This ordering reveals a separation between *representational quality* and *task performance*: while Mod A+B produces the "best" representations by this invariance metric, it does not achieve the highest reward, suggesting that excessive invariance may discard useful task-relevant information.

t-SNE visualization. Figure 4 shows t-SNE projections colored by episode return. The baseline distractor plot shows a wide return spread (770–855) with scattered clusters, indicating inconsistent behavior. In contrast, Mod B distractor exhibits a tight return range (790–796) with well-organized structure, demonstrating that the task-conditional contrastive loss produces the most *behaviorally coherent* representations—even though it suffers from training instability.

5.5 Computational Overhead

Table 3 reports wall-clock time per training step, averaged over 1000 updates.

Table 3: Computational overhead per training step (RTX 5090 Laptop GPU).

Method	**ms/step**	**Overhead vs. baseline**
Baseline	48.3	—
+ Mod A	45.8	≈0%
+ Mod B	60.3	+25%
+ Mod A+B	95.0	+97%

Mod A adds **zero overhead** (the 2.5ms speedup is within measurement noise, likely due to GPU batching effects). This is because DrQ-v2 already computes two augmented views for the actor and critic updates; Mod A reuses one of these and only adds an MSE loss computation. Mod B adds 25% overhead from pairwise Q-value comparison and InfoNCE computation. The combination nearly doubles training time, making it impractical for resource-constrained settings.

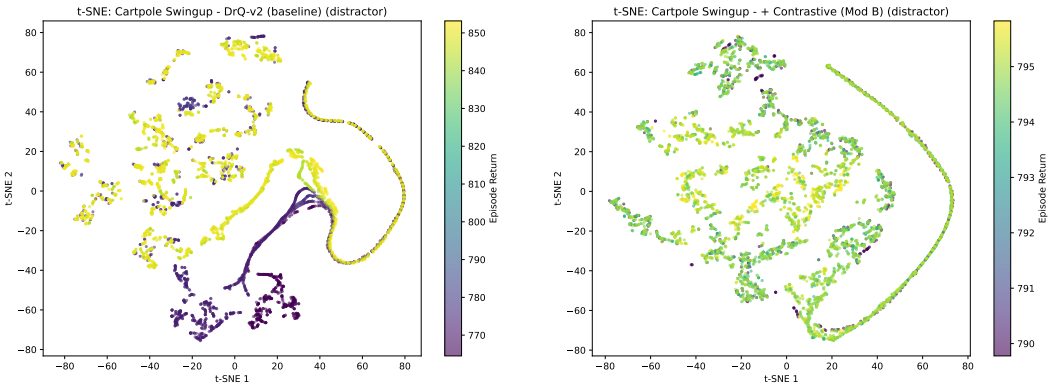
6 Discussion

Mod A is the recommended practical improvement. Across all evaluation axes—distractor robustness, cross-seed stability, α-sensitivity, effective rank, and computational

(a) Effective rank across methods.

(b) CKA: clean vs. distractor.

Figure 3: Representation analysis (cartpole, seed=1). (a) Mod A achieves highest effective rank (22.5), indicating fuller feature utilization. (b) CKA heatmap: Mod A+B produces the most invariant representations (0.78 on-diagonal), but the baseline is most disrupted (0.54).



(a) Baseline (distractor): wide return spread.

(b) Mod B (distractor): tight, organized returns.

Figure 4: t-SNE of latent representations colored by episode return. The baseline exhibits scattered, return-diverse clusters (770–855), while Mod B produces behaviorally coherent representations with minimal return variation (790–796).

cost—Mod A emerges as the most practical improvement to DrQ-v2. It achieves the highest distractor reward with the smallest variance, is insensitive to the loss weight across two orders of magnitude, and adds zero computational overhead. For practitioners deploying DrQ-v2 in visually complex environments, adding Mod A with $\alpha = 0.1$ is a "free lunch."

The representation–performance paradox. Our CKA analysis reveals a surprising disconnect: Mod A+B produces the most distractor-invariant representations (CKA $= 0.78$) but does not achieve the highest reward. This suggests that **representation invariance and task performance are not monotonically related**. Excessive invariance may discard task-relevant spatial information that the critic needs. This finding echoes concerns

raised by Zhang et al. (2021) about the trade-off between invariance and informativeness in bisimulation-based methods.

Task-conditional contrastive losses are fragile in RL. Mod B's instability—seed-dependent collapse in clean cartpole, non-monotonic α-sensitivity, and catastrophic failure on walker distractors (reward 26.6)—reveals a fundamental challenge in coupling contrastive losses with RL critics. The Q-value-based pairing mechanism creates a **circular dependency**: accurate positive pairs require good Q-estimates, but good Q-estimates require a well-trained encoder, which is exactly what the contrastive loss is supposed to provide. When this bootstrap fails (due to seed-dependent initialization, α-induced gradient scaling, or distractor-corrupted Q-estimates), the system enters a collapse attractor. The walker distractor result is particularly instructive: while visual distractors *stabilize* Mod B on cartpole (all 3 seeds succeed), they *destroy* it on walker, suggesting that the interaction between task complexity, observation dimensionality, and contrastive learning is highly non-trivial.

No universal auxiliary loss. The cross-task results (Table 2) demonstrate that the optimal auxiliary loss is strongly task- and environment-dependent: Mod A excels on walker and cartpole distractors, Mod B excels on cheetah clean but catastrophically fails on walker distractors, and the baseline itself proves surprisingly robust on walker. This motivates future work on **adaptive loss selection** or **dynamic weighting** that adjusts the auxiliary loss based on training signals.

Limitations. (1) Cartpole is the only task with full 3-seed evaluation; walker and cheetah use a single seed, preventing statistical significance testing for these tasks. (2) Our distractor backgrounds are synthetically generated; real-world videos (e.g., Kinetics) may present qualitatively different challenges. (3) We evaluate at fixed training budgets (500K and 1M frames) and do not study asymptotic performance. (4) The Q-value threshold $\epsilon = 5.0$ in Mod B was not extensively tuned and may be suboptimal for tasks with different reward scales.

7 Conclusion

We presented a systematic evaluation of two lightweight auxiliary representation losses for improving DrQ-v2's robustness to visual distractors. Our experiments across three DMC tasks demonstrate that:

1. **DrQ-v2 is vulnerable to distractors**: the baseline drops $\sim$11% under video backgrounds on cartpole, confirming the vulnerability identified in our Assignment 1 analysis.
2. **Mod A (consistency regularizer) is the most practical improvement**: it achieves the highest distractor reward (782.6$\pm$7.8), smallest variance, zero overhead, and robust α-insensitivity. We recommend adding it to DrQ-v2 as a default.
3. **Mod B (contrastive loss) produces the best representations but is dangerously unstable**: it yields the tightest t-SNE clusters and strong CKA scores, but 1/3 clean seeds collapse, 1/4 α values fail, and it catastrophically fails on walker distractors (reward 26.6), revealing a fundamental fragility in Q-value-coupled contrastive learning.
4. **Representation invariance $\neq$ task performance**: the combined method achieves the highest CKA (0.78) but not the highest reward, suggesting that blindly maximizing invariance is suboptimal.

Future work should explore adaptive loss weighting (e.g., scaling α based on Q-value confidence), warm-starting Mod B after initial Q-function convergence, and evaluation on high-dimensional manipulation tasks with real-world visual complexity.

8 Team Contributions

- **Bangcheng Wang**: Implemented Modification A and B in the DrQ-v2 codebase; designed and ran α-sensitivity ablation experiments; wrote Methodology, Analysis, and Discussion sections of the report.
- **Kean Chen**: Set up the DMC-GB distractor environments with video backgrounds; ran main comparison experiments (clean and distractor, all seeds); conducted representation analysis (CKA, effective rank, t-SNE); wrote Experiments, Results, and Figures sections.
- **Shared**: Introduction, Related Work, Conclusion, Presentation preparation, code cleanup.

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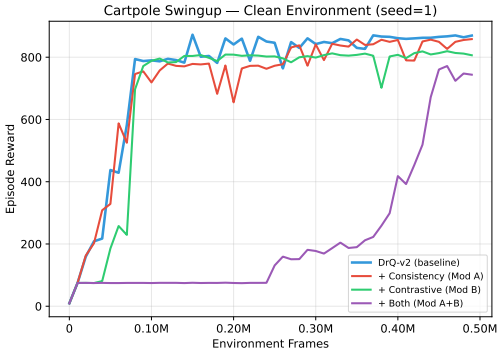
A AI Declaration

This project was developed with assistance from the following AI tools:

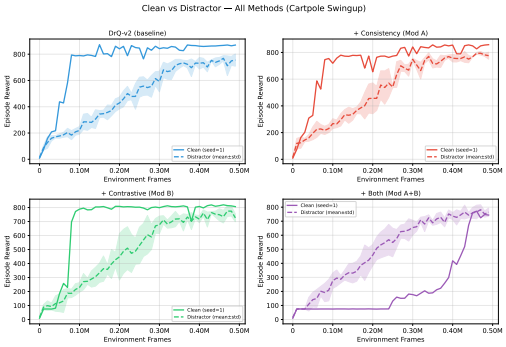
- **Claude (Anthropic)**: Used for code implementation assistance (auxiliary loss modules, distractor environment setup, experiment automation scripts), debugging, data analysis pipeline, and LaTeX report drafting.
- **Gemini (Google)**: Used during the proposal phase for literature screening and preliminary synthesis of related works.

All experimental results are from actual training runs conducted on the authors' hardware. AI-generated code was reviewed, tested, and validated by the authors. The main research hypotheses, experimental design, and analysis of results were developed by the authors with limited AI assistance, in accordance with the course AI policy. All AI-generated text was reviewed and edited by the authors for technical accuracy and academic integrity.

B Additional Figures

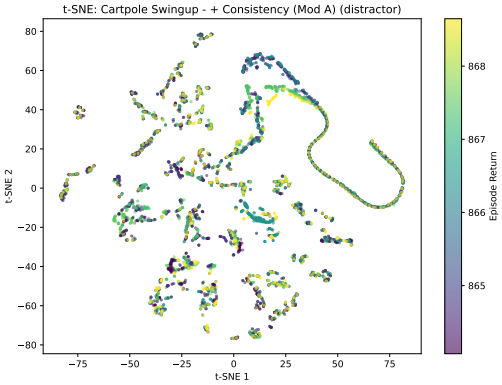


(a) Clean environment learning curves (3 seeds). All methods converge to similar performance, with Mod B showing higher variance due to one collapsed seed.

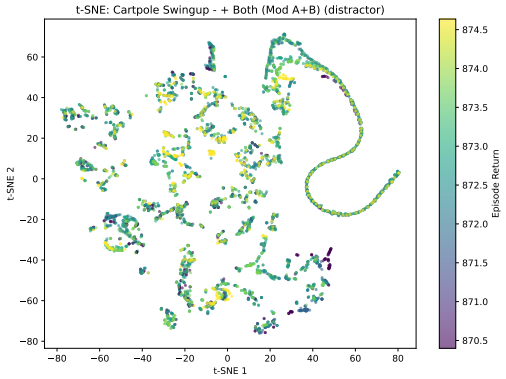


(b) Direct clean vs. distractor comparison for each method. Mod A shows the smallest gap; baseline shows the largest.

Figure 5: Additional cartpole swingup results.



(a) Mod A (distractor)



(b) Mod A+B (distractor)

Figure 6: Additional t-SNE visualizations under distractors (cartpole, seed=1).